SHREYAS ACHARYA

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SKILLS

Languages: Python, C++, SQL, JavaScript, MATLAB
Deep Learning & Robotics: Reinforcement Learning, Imitation Learning, Robotic Manipulation, Planning, Model Alignment, Motion Planning, Multimodal Foundation Models, Generative Models (Diffusion/GANs), Deep Learning, Computer Vision
Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Scikit-Learn, CUDA, Open3D, Stable-Baselines3, Hugging Face, TensorRT, React
MLOps, Tools & Simulation: Git, GitHub Actions, Docker, RESTful APIs, CI/CD, MLflow, CMake, AWS, Distributed Deep Learning, PyBullet, MuJoCo

EXPERIENCE

Williams-Sonoma, Inc. — **Nov 2025 – Present**
AI Research Engineer — *San Jose, CA*

- Building Generative AI and Deep learning algorithms to process millions of images for custom asset generation and quality control.

Perception Robotics Group — **Jan 2024 – June 2024**
Graduate Research Assistant — *College Park, MD*

- Achieved **99%** 3D reconstruction accuracy from raw tactile sensor data as measured by IoU and point cloud density, by developing an end-to-end Reinforcement Learning pipeline featuring a novel CNN-based policy network and temporal stacking.
- Increased object reconstruction robustness by over **15%** across various complex geometries as measured by policy performance albation studies, by engineering a multi-term RL reward structure that balanced exploration with area maximization.
- Reduced training time by 40% through optimized sensor movement and simulation environment, maintaining reconstruction quality and enabling efficient tactile exploration.

PROJECTS

Pb-lite Edge Detection: An Interactive Web Application — **July 2025**
Python, OpenCV, Scikit-learn, SciPy, FastAPI, Docker

- Engineered a full-stack web application by integrating FastAPI, achieving a 99.9% uptime and reducing API response latency by 25% through optimized caching, rate limiting, and resource management.
- Implemented edge detection pipeline, processing high-resolution images 2x faster than traditional methods by leveraging filter bank caching and efficient in-memory operations, enabling real-time edge detection for users.
- Deployed a containerized backend on Hugging Face using Docker, reducing deployment time by 30% and ensuring scalability with a lightweight image, while integrating Google Analytics for actionable insights into user behavior.

Temporal Coherence Evaluation for Multimodal Foundation Models — **Nov 2024**
Python, PyTorch, Hugging Face, CLIP

- Conducted comprehensive research assessing temporal reasoning in state-of-the-art multimodal foundation models (Vision-language models) in PyTorch using semantic similarity and a novel reference-free metric(CLIPGain).
- Benchmarked six multimodal foundation models, revealing up to 14× improvement in cross-frame consistency for long-context architectures.

Multiview Structure from Motion for 3D Scene Reconstruction — **May 2024**
Python, OpenCV, NumPy, SciPy, Open3D

- Designed a 3D reconstruction pipeline to accurately reconstruct scenes from multiple images, employing SIFT/FLANN-based matching for enhanced accuracy across synthetic and real-world datasets.
- Incorporated custom Bundle Adjustment optimization to jointly refine 3D points, reducing reprojection errors by 10x across all datasets while preserving structural integrity, yielding high-fidelity 3D structures.

Live Human Activity Recognition System — **June 2023**
Python, TensorFlow, OpenCV, Scikit-learn, SciPy, FlaskAPI, ONNX

- Architected and deployed a CNN-based real-time Human Action Recognition System, achieving 98% accuracy at 30 frames per second using a consecutive frame validation while supporting remote monitoring and automated alert logging.

- Developed a hybrid multi-threaded architecture combining a desktop GUI with a Flask web interface, supporting concurrent monitoring of video streams while reducing latency by 40% through an optimized ONNX inference model.
- Developed an intelligent action monitoring and alert system to raise context-aware alerts, incorporating temporal analysis that improved detection accuracy by 24%, enabling real-time logging and intervention capabilities.

Flavors Academy: A modern Learning Management System **Nov 2024**
React, python, SQL, Docker

- Led architecture development of a full-stack learning management system with 3 user profiles using React and Python with RESTful APIs. Collaborated in an agile environment to deliver a scalable software solution.
- Developed automated tests using pytest to ensure system reliability, containerized the application with Docker for a scalable, production-ready deployment.

Dynamic RRT* Path Planning with Real-time Obstacle Avoidance **May 2024**
Python, PyGame, NumPy, Priority Queue

- Achieved 100% collision-free navigation in complex, dynamic environments as measured by exhaustive simulation testing, by pioneering a dynamic RRT path planning algorithm with real-time obstacle avoidance.
- Reduced path planning computation time by 40% measured by sub-second replanning times, by implementing optimized node pruning and efficient tree rewiring algorithms crucial for real-time robotic controls

Adaptive A* Path Planning System for Autonomous Navigation **April 2024**
Python, OpenCV, NumPy, ROS2

- Implemented A* path planning algorithm in Python with custom obstacle inflation and clearance handling, achieving 100% success rate in navigating complex 2D environments.
- Developed a navigation system supporting a TurtleBot3 in Gazebo simulation environment integrating with ROS2 navigation stack, enabling real-time adaptive path planning.

Homography-Net: Deep Learning-Based Image Alignment & Panorama Stitching **March 2024**
Python, TensorFlow, NumPy, Keras

- Designed a deep neural network (VGG-19 backbone) with optimized supervised and unsupervised pipelines to estimate homography transformations between image pairs, achieving sub-5 pixel error and reducing prediction error by 68%.
- Built a robust vision-based image stitching system using Shi-Tomasi corner detection and RANSAC to align images with 98.4% feature match accuracy, scalable for large visual datasets.

Autonomous Path Planning System with Dijkstra's Algorithm **March 2024**
Python, OpenCV, NumPy, Priority Queue

- Reduced path planning computation time by 60% through the implementation of optimized Dijkstra's algorithm with a priority queue and early goal detection.
- Increased system reliability to a 100% success rate in obstacle avoidance through robust configuration space handling and safety margin implementation.
- Accelerated development cycle by 80% through an automated visualization and testing pipeline with real-time performance metrics.

Pb-lite Edge Detection: Multi-Cue Gradient-Based Boundary Detection **Jan 2024**
Python, TensorFlow, Scikit-learn, SciPy

- Developed an enhanced edge detection algorithm to accurately detect edges by combining brightness, color, and texture gradients across multiple scales, inspired by the classical Pb (probability of boundary) approach.
- Improved edge continuity by approximately 30% through iterative refinement, significantly enhancing the detection of fine structures and preserving object boundaries with high spatial fidelity.

EDUCATION

University of Maryland, College Park **College Park, MD**
Master of Engineering in Robotics *May 2025*
Coursework: Multimodal Foundation Models, Computer Processing of Pictorial Information, Robot Learning, Data Structures and Algorithms

Savitribai Phule Pune University **Pune, IN**
Bachelor of Engineering in Computer Engineering *May 2023*